SECURITY ENGINEERING RESEARCH POSTER

Statistical Screening for Poisoned Machine-Learning Training Data

Detection and Quarantine of Suspicious Samples at the Data-Ingestion Boundary

github.com/poojakira/dataset-poisoning-detector - public, default branch main.

EVIDENCE POSTURE

Verified, scoped, reproducible

Claims are constrained to the repository evidence snapshot and stated limitations.

226	186	91.75%	0.55
Tests passed (includes 40 legacy stub-only tests)	Production-facing tests passed	Statement coverage	Cross-class F1 @ 5% poison

SYSTEM VIEW

Architecture and Threat-Control Flow

Trust Boundary

Input surface
Data, models, requests, policies, or artifacts entering the tested path.

Control surface
Repository-specific validation, analysis, and fail-closed checks.

Evidence surface
Findings, metrics, tests, logs, or reproducible artifacts.

This diagram describes the tested repository path, not a universal deployment guarantee.

01 **Validate incoming feature batches and streaming samples.**

02 **Apply z-score, IQR, Isolation Forest, and spectral/label-aware detectors.**

03 **Combine detector signals for screening and quarantine workflows.**

04 **Maintain streaming state and deduplication metadata.**

05 **Benchmark label-flip behavior with committed reproducible artifacts.**

Research Contribution

An ingestion-boundary screening pipeline combining statistical outlier methods with a label-aware cross-class centroid detector that improves the committed synthetic label-flip benchmark from the old top-1 spectral baseline 0.08 / 0.23 / 0.3675 F1 to 0.55 / 0.595 / 0.6975 F1 at 5% / 10% / 20% contamination.

Verified Evidence

EV-01	Cross-class F1 @ 10% poison: 0.595	EV-02	Cross-class F1 @ 20% poison: 0.6975
EV-03	Old top-1 spectral F1: 0.08 / 0.23 / 0.3675	EV-04	Feature-space ensemble F1: 0.0841 / 0.1421 / 0.2331
EV-05	226 tests passed, 0 skipped (186 production-facing and 40 legacy demonstration/stub tests).	EV-06	91.75% statement coverage; CI gate is 90%.
EV-07	Ruff check/format and Pyright pass on the final branch; GitHub currently reports 0 open Code Scanning, Dependabot, and secret-scanning alerts.	EV-08	Committed results/spectral_benchmark.json records cross-class centroid F1:
EV-09	0.55 at 5% contamination	EV-10	0.595 at 10%

Method and Control Path

01 Validate incoming feature batches and streaming samples.

02 Apply z-score, IQR, Isolation Forest, and spectral/label-aware detectors.

03 Combine detector signals for screening and quarantine workflows.

04 Maintain streaming state and deduplication metadata.

05 Benchmark label-flip behavior with committed reproducible artifacts.

Limitations and Residual Risk

- The benchmark is synthetic and separable; it is not proof of production detection rate.
- Clean-label, subtle backdoor, and very-low-rate poisoning remain difficult.
- Throughput is hardware/environment dependent; no fixed product SLA is claimed.
- Stateful streaming behavior requires deliberate partitioning/external state for horizontal scaling.

Reproducibility

```
Expected current CI evidence: 226 passed, 91.75% coverage
(includes 40 legacy stub tests). Measured code snapshot:
45740923aa16539909a7ae993104a80ecfe85e0b Evidence CI run:
38502623976
```

Evidence Provenance

Claim discipline
Metrics come from the repository's verified metrics and research brief.

Scope discipline
Benchmarks are reported only at their documented fixture, dataset, or test scope.

Reproduction
PDF and PNG are generated locally from committed evidence without paid services.

Evidence-backed security engineering

README, verified metrics, tests, CI, and committed artifacts remain authoritative. Poster layout emphasizes architecture, controls, evidence, limitations, and reproduction rather than unsupported efficacy claims.

36 x 48 in
PDF + PNG parity